

Non repudiation

Non repudiation is a provision of Digital signatures to ensure that the signer cannot claim that they never signed a document

Secure signatures

To enable usage of a digital signature on multiple computers, it can be securely stored on a smart card

encryption garbles the content of the message / document / data in such a way that only the intended parties can decode it.

Encryption gives security to the content by making it impossible for anyone to access it unless they specifically know how. If digital signatures are similar to putting a name tag on your lunch, and authentication tags are like wrapping it in plastic, then encrypting is analogous to locking it in a 10-tonne stainless steel safe.

Encryption relies on algorithms for which the coding is far simpler than the decoding. The encoding operates on two things, the data, and the key. The algorithm is designed such that without the key it is nearly impossible (or unfeasible) to get the data back from the cypher. However unless combined with a digital signature mechanism, you have no way of knowing if the encrypted data is from right source.

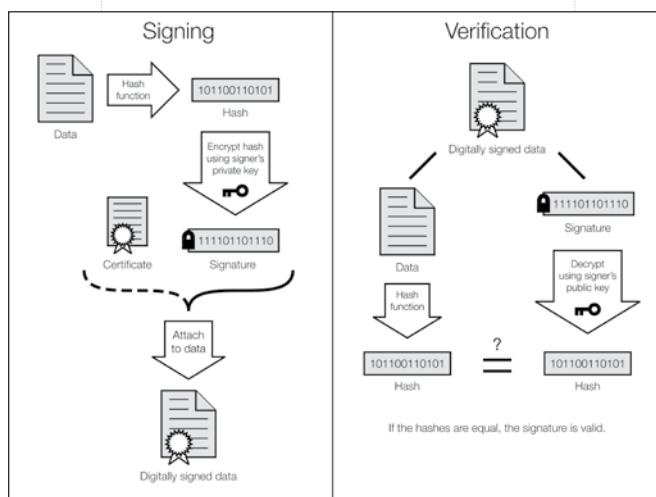
Encryption schemes also present another conundrum. How do you transfer the encryption "secret"? Whether it is an algorithm or a key, transferring it through an insecure medium is as good as not encrypting at all.

Private-Key encryption

The private key encryption method is one that most people are aware of. Two parties agree on a secret key which will be used to encrypting any further communication. This initial communication where the key is exchanged can be done in person or via a secure channel. Much like whispering or passing a note.

The obvious disadvantage here is the need for a secure way of communication, and if that is available, the whole need for encryption becomes a little moot.

In simple cases, we may share a password over telephone with our friend for files we are about to share. When it is your job to be paranoid about your companies security however, that doesn't cut it, and a security enabled channel will be used. If having a secure communication channel is expensive, we can use the channel merely to exchange the key. After which the internet can be host to further communications which



Signing a digital document with a "hash" key is a common security measure

have been encrypted with the private key.

Public-Key encryption

Unlike the public key system, which is symmetric, i.e. the encrypting key and decrypting key are the same; public-key encryption uses different keys for encrypting and decrypting data. This may seem counter-intuitive, but is incredibly useful. In a Public-Key encryption, the encryption key is openly published for anyone to see, while the decryption key is kept secret. What this essentially means is that while everyone is capable of encrypting messages that you can decrypt, however they are unable to decrypt any messages encrypted by the public-key even the ones they encrypted themselves!

In this case, a secure line of communication can be obtained by having the participants exchange public keys, and use them for any secure communication. A secure channel is not required for sending the keys in this case, since the key is only capable of encrypting and not decrypting, furthermore, it is impractical to determine the decrypting key given the encrypting key. Even if your adversaries were to gain access to your public key, it would only enable them to encrypt data with your key, they would still be helpless to decrypt any data that they intercept.

In this kind of system, the sending party encrypts confidential data with the public-key of the party which is to receive said data. The recipient of the data is then able to decrypt the message

since they have a decryption key (their private-key).

When we use digital signatures practically, a software generates a signature for our given message. The signature is such that if any part of the message / document is changes, the signature would no longer match and we would know immediately that someone has tampered with the message.

Digital signatures aren't limited to email messages or documents, they are not used in almost every field where the source of the data is important. Microsoft uses digitally signed certificates for drivers it has tested, and warns the user if an unsigned driver is being installed. Digitally signed files, if modified by a virus, will invalidate the signature, which can serve a warning to the

user.

A signature is only as good as the one who signed it. A million dollar cheque will do you little good if the signature is of a pauper. Likewise, a digital signature by its nature needs to be easy to produce and as such by itself shouldn't inspire any confidence. A digitally signed virus which deletes all your files, will only do its job if its signature is intact.

For simple cases of authenticating your emails and files for personal purposes, digital signatures are almost free to use. There are many tools capable of generating signature for your messages and documents. Software that work with documents have some support for digital signatures. Microsoft Office products, for example, allow you to generate and attach a digital signature to your documents.

Conclusion

As we see our digital identities take a increasingly important role in our lives, we need to protect it as much as we can. With digital signatures and encryption we can ensure that our message gets across unhindered and without misrepresentation. The need for security is thus becoming more apparent. The measures of security and privacy are becoming a more integral part of every online service.

The internet is no longer just for communicating with people far away. We use it daily to communicate with people in our schools, offices, with our neighbours or even the people in our own homes. This makes what we do online even more important and relevant to our daily lives. There is no longer a Neo and Mr. Anderson, we now have one personality of which the internet is a part.

The ease with which we can create online identities for ourselves only exacerbates the fact that it is as easy for someone else to assume our identity online as it is for us. 



Digitally sign your emails

It is quite easy to use digital signatures with Thunderbird. For this one merely needs to install an addon called Enigmail and an OpenPGP system such as GnuPG or gpg4win, all of which are free and open source software.

After installing both of them, you will see an OpenPGP menu in thunderbird. Enigmail comes with a setup wizard located in the OpenPGP menu, which will allow you to easily set up your email client to use the installed openPGP software to add digital signatures to your email.